

SECTIONS

Headline
Geo-spatial bias
Per-agent metrics
Judge scores
Hallucination

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VLM Council — Hub-and-Spoke Evaluation

COUNTRY ACCURACY

64.6%

MEDIAN HAVERSINE

435 km

IMAGES

500

MEAN DISCUSSION
ROUNDS

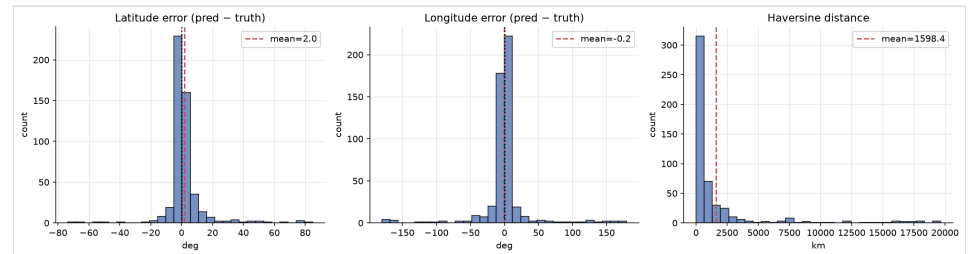
1.31

MEAN JUDGE STRATEGY

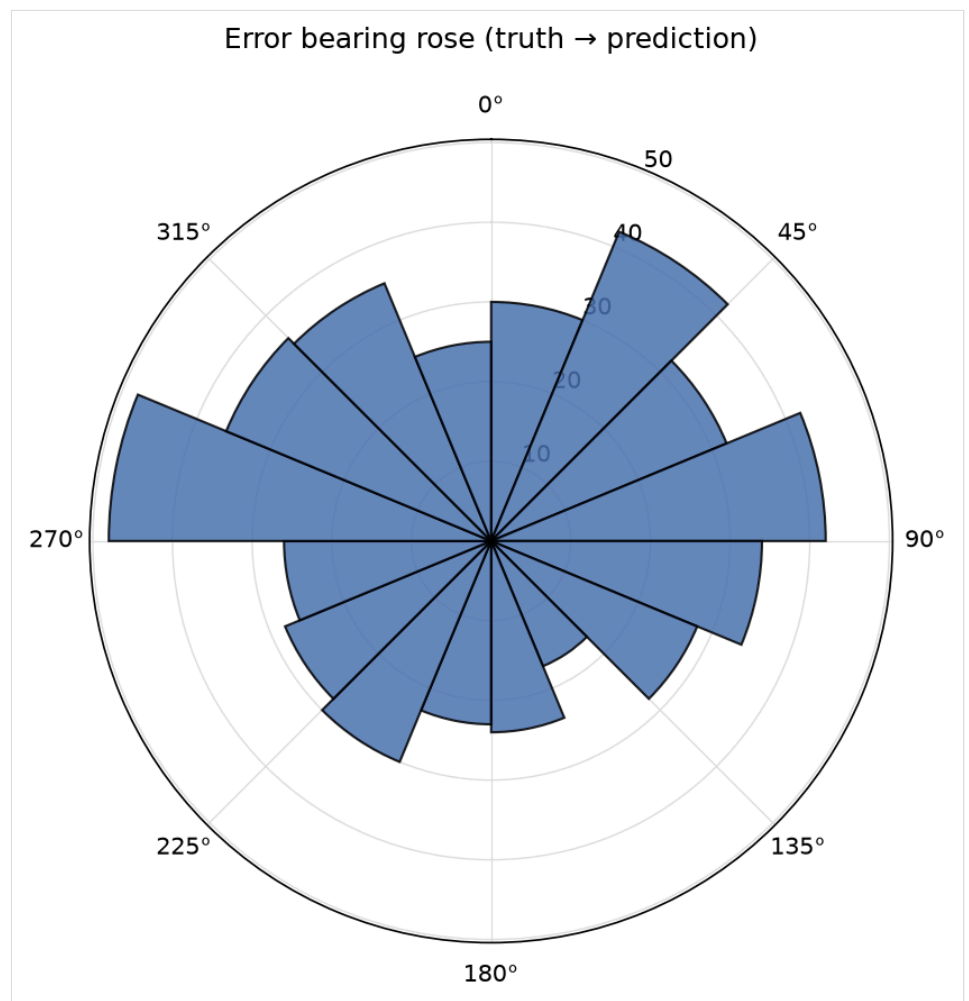
0.58

Geo-spatial bias

- North/south bias: strong north bias ($p=0.0005$)
- East/west bias: no significant bias ($p=0.8940$)
- Error quadrants: SW=116 NE=142 NW=135 SE=107

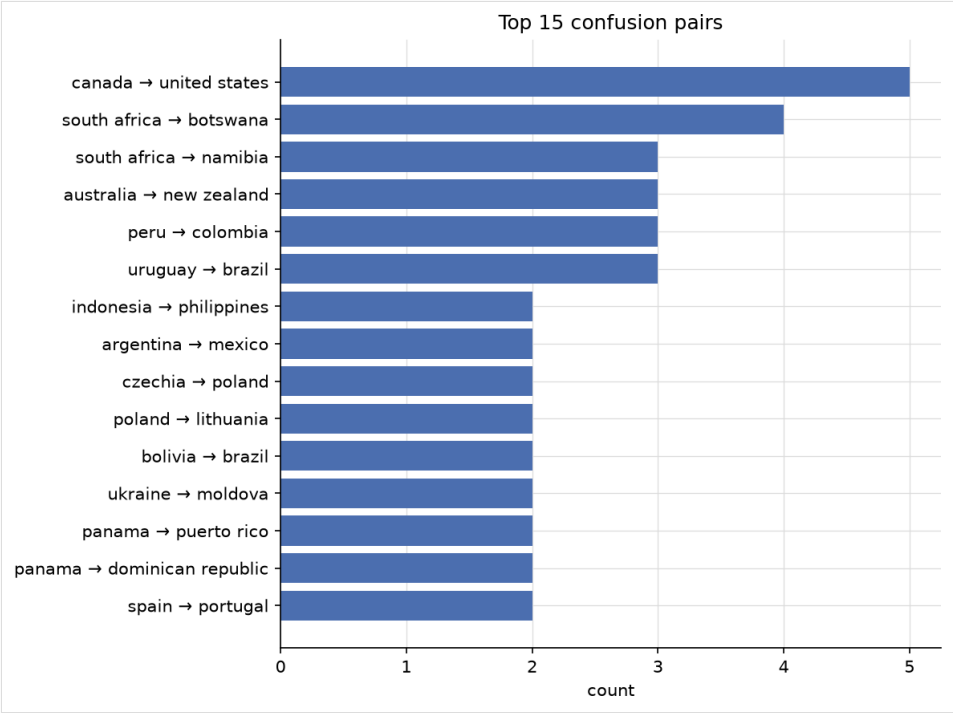


Lat/lon error distributions and haversine distance.



Bearing of prediction errors (truth → prediction).

► Top confusion pairs (20)

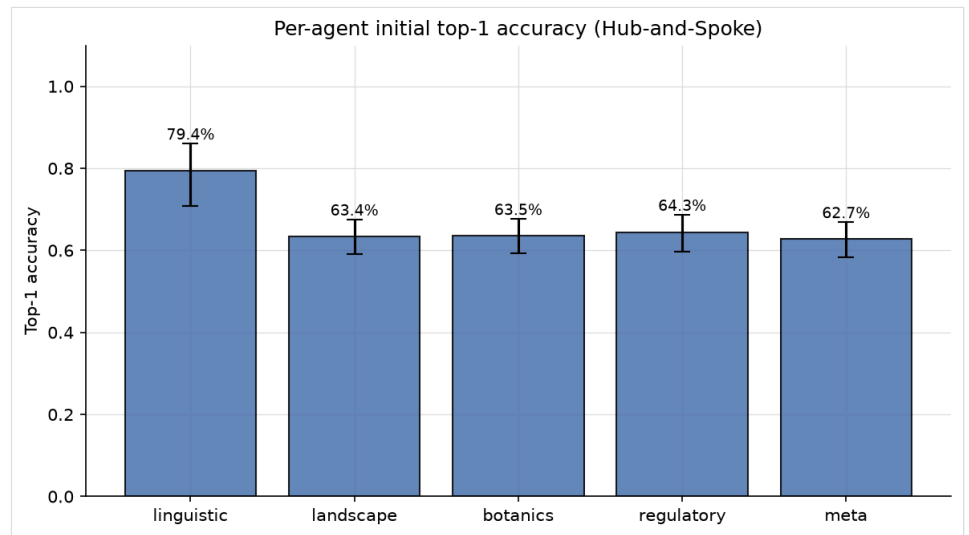


Top country confusion pairs.

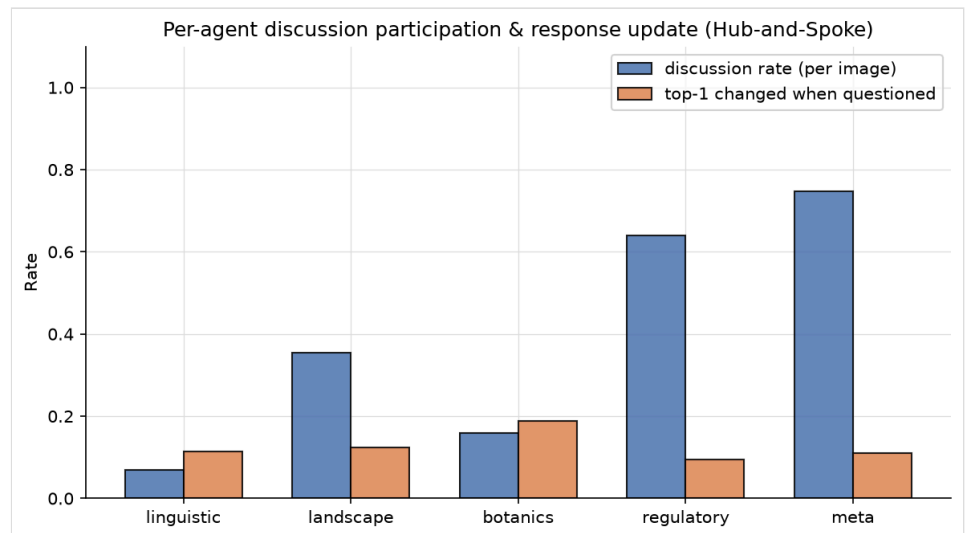
Per-agent metrics

Initial-round accuracy and Hub discussion participation statistics. **Discussion rate** = fraction of images where this agent was questioned by the judge hub. **Update rate** = fraction of times agent changed top-1 country when questioned.

Agent	n	Top-1	Top-3	Coverage	Discussion rate	Update rate
linguistic	107	79.4%	86.0%	90.7%	7.0%	11.4%
landscape	500	63.4%	80.0%	82.2%	35.4%	12.4%
botanics	499	63.5%	79.0%	79.8%	16.0%	18.8%
regulatory	443	64.3%	77.4%	78.3%	64.0%	9.4%
meta	499	62.7%	75.2%	75.2%	74.8%	11.0%



Per-agent initial top-1 accuracy with Wilson 95% CI.



Per-agent discussion participation rate and top-1 update rate when questioned.

LLM-as-judge scores

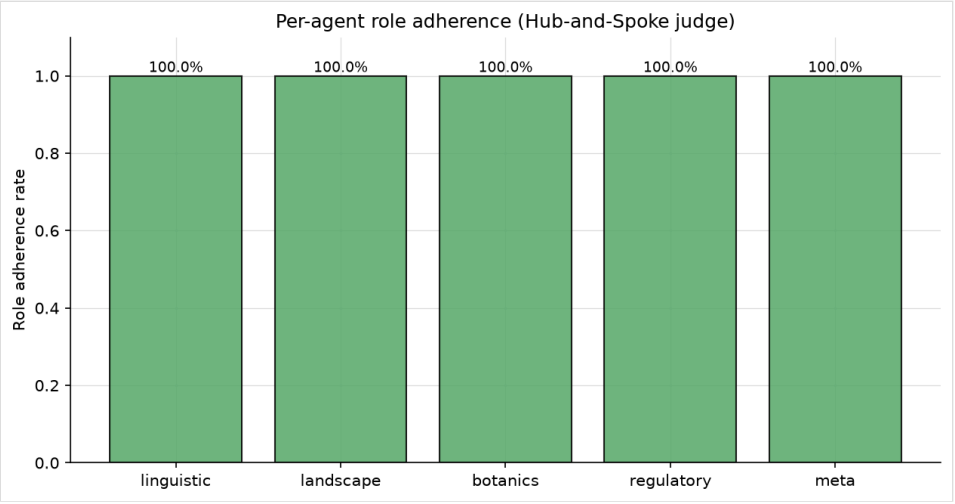
Verdicts: 500 / 500

Judge question strategy score: mean 0.576, median 0.500, σ 0.278 (n=500)

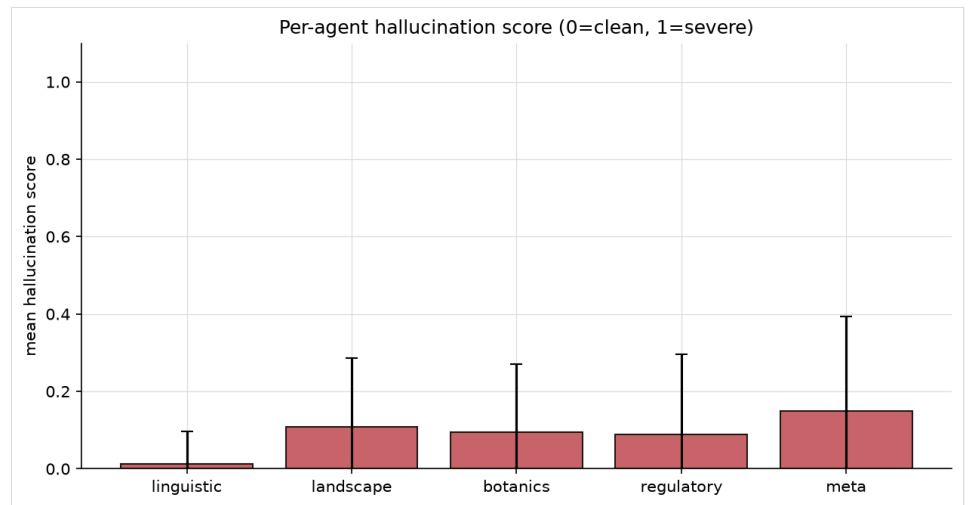
Judge synthesis quality: mean 0.623, median 0.750 (n=500)

Per-agent scores

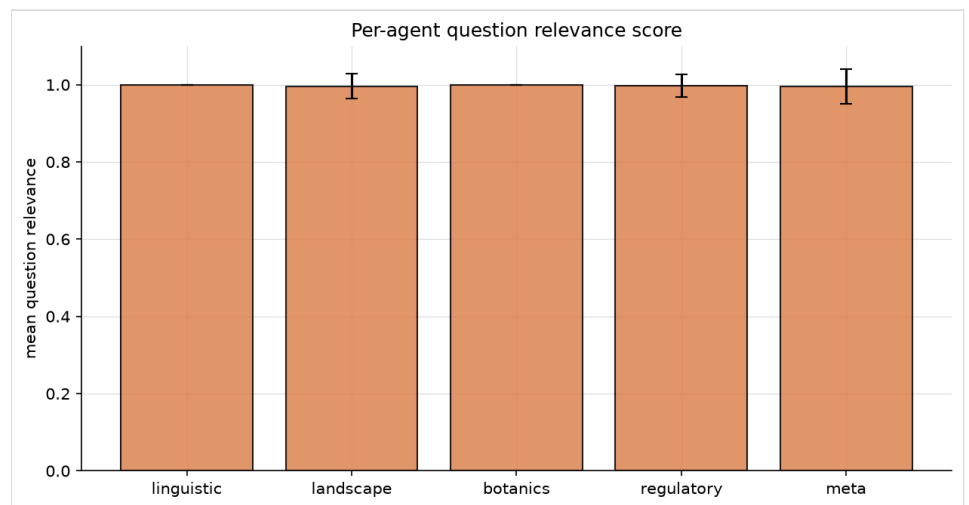
Agent	n	Role adher.	Halluc. ↓	Visual cons. ↑	Calib. ↑	Q- relevance ↑	Resp. update ↑
linguistic	500	100.0%	0.01 ± 0.08	0.99 ± 0.06	0.58 ± 0.20	1.00 ± 0.00	0.87 ± 0.25
landscape	500	100.0%	0.11 ± 0.18	0.90 ± 0.15	0.65 ± 0.25	1.00 ± 0.03	0.74 ± 0.26
botanics	500	100.0%	0.09 ± 0.18	0.89 ± 0.16	0.64 ± 0.24	1.00 ± 0.00	0.72 ± 0.26
regulatory	500	100.0%	0.09 ± 0.21	0.92 ± 0.20	0.63 ± 0.25	1.00 ± 0.03	0.67 ± 0.30
meta	500	100.0%	0.15 ± 0.24	0.85 ± 0.23	0.62 ± 0.26	1.00 ± 0.04	0.69 ± 0.30



Per-agent role adherence rate.



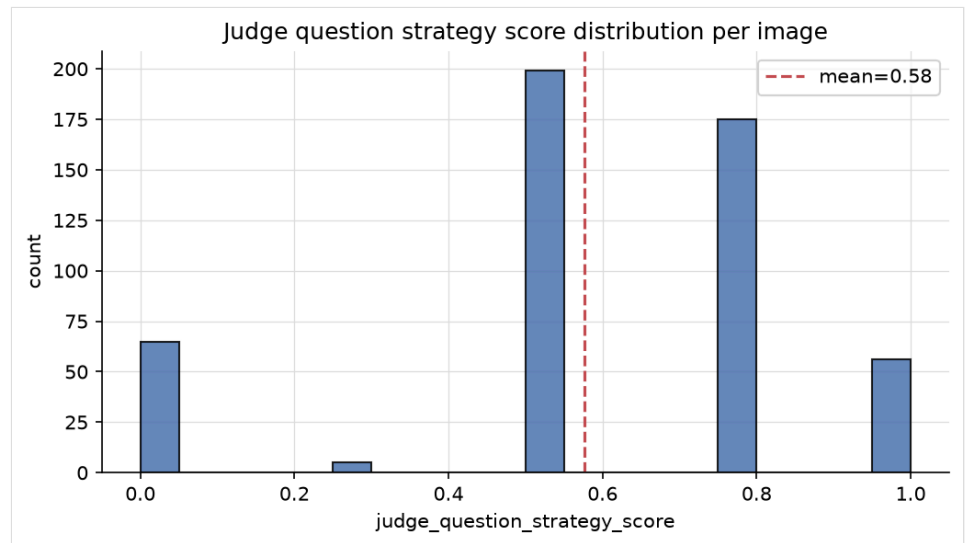
Per-agent mean hallucination score (0 = clean, 1 = severe).



Per-agent question relevance score (how in-domain was the judge's question?).



Per-agent response update quality (new information when questioned?).



Distribution of judge question strategy scores per image.

Hallucination examples

Concrete claims the judge flagged as not supported by the image.

► linguistic — 10 flagged claim(s)

► landscape — 10 flagged claim(s)

► botanics — 10 flagged claim(s)

► regulatory — 10 flagged claim(s)

► meta — 10 flagged claim(s)